

Timber workshops — ground-screw foundation & floor plan

HEAVY-DUTY — 1.5 m bearer spacing, joist spans ≤ 1.5 m

Up-rated layout for heavier floor loads — machinery, a vehicle, dense storage. Support lines are tightened from 2.0 m to 1.5 m, adding one bearer line (and its screws) per room. Same 5"×2" timber throughout. Prepared 30 Jul 2026.

Standard vs heavy-duty at a glance

	Standard (2.0 m spans)	Heavy-duty (1.5 m spans)
Bearer lines	WS1: 3 WS2: 3	WS1: 4 WS2: 4
Ground screws	~33 total	~43 total
5×2 timber (with waste)	~240 lin. m	~260 lin. m
Best for	general / light workshop use	vehicles, machinery, heavy storage

Design assumptions

- **Joists:** 5"×2" (47×125 mm), ideally C24 graded, at **400 mm centres**.
- **Max joist span 1.5 m** between supports — a comfortable margin for heavy point loads.
- Joists sit on **5×2 bearer beams** at ≤ 1.5 m centres, which sit on the ground-screw heads.
- Ground screws with **adjustable U-head / plate tops**, at ≤ 1.5 m centres along each bearer.
- Deck on top (not in this take-off): 18–22 mm OSB3 T&G.

Headline quantities

- **Ground screws:** **~43 total** — 20 for Workshop 1, ~23 for Workshop 2 (order ~46 with spares).
- **5×2 timber:** **~235 linear metres** of framing — call it ~260 m allowing for waste & noggins.

Workshop 1 — 6.0 × 4.0 m

- **Ground screws:** **20** — 5 columns ($X = 0, 1.5, 3.0, 4.5, 6.0$ m) × 4 bearer lines ($Y = 0, 1.33, 2.67, 4.0$ m).
- **Bearers:** 4 runs × 6.0 m = **24 m**.
- **Joists:** 16 × 4.0 m (400 c/c) = **64 m**.
- **Noggins** over the 4 bearer lines: **24 m**. Room total 5×2 ≈ **112 m**.

Workshop 2 — ~4.0 × 7.2 m (clipped corner)

- **Ground screws:** **~23** — 4 bearer lines at $X = 0, 1.33, 2.67, 4.0$ m; screws at $Y = 0, 1.5, 3.0, 4.5, 6.0, 7.2$ m. The two left lines are cut short by the angled wall, plus 2 extra at the door threshold.
- **Bearers:** ≈ **27 m** (left runs shortened by the clip).
- **Joists:** ~19 × 4.0 m ≈ **72–76 m**.
- **Noggins:** **~26 m**. Room total 5×2 ≈ **125 m**.

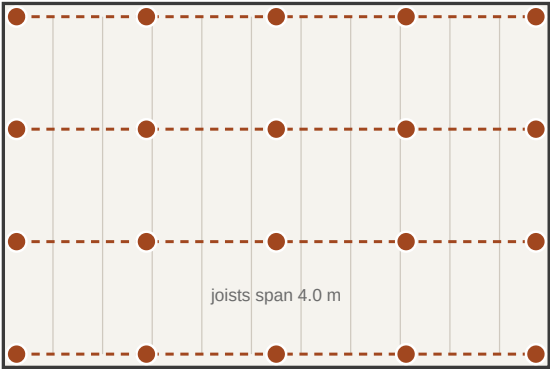
Shopping list

Item	Quantity	Notes
5×2 joists	35 × 4.2 m lengths	same as standard (joist count unchanged)
5×2 bearers + noggins	~22 × 4.8 m lengths	the extra bearer line per room lands here

5×2 total	≈ 57 lengths	~260 lin. m incl. waste
Ground screws	46 (43 + 3 spares)	adjustable U-head / plate top

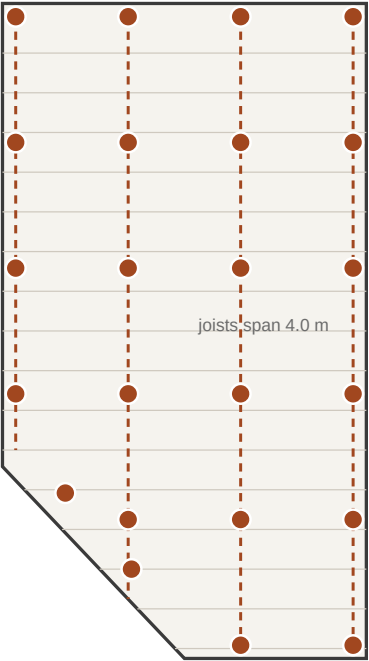
Ground-screw & joist layout

Workshop 1 - 6.0 x 4.0 m - 20 ground screws



- Ground screw - 43 total (buy ~46 with spares)
- - - 5x2 bearer @ 1.5 m centres on adjustable screw heads
- 5x2 joists @ 400 mm centres (span ≤ 1.5 m)

Workshop 2 - ~4.0 x 7.2 m - 23 ground screws



clipped corner = angled door wall

Notes & alternatives

- **Alternative — keep 6×2 joists at the standard 2.0 m layout.** That holds screws at ~33 (fewer holes to dig) but uses deeper, pricier timber. This sheet takes the other route: stay on 5×2, add support lines.
- **Screw length depends on the ground, not the building.** Heavier loads make firm founding depth more important — match the supplier load ratings to the 1.5 m spacing.
- **Confirm Workshop 2's exact shape.** The clip is estimated to hit the 24.46 m² internal area — give the true corner points and the WS2 counts can be tightened.

Indicative take-off for planning only — not a structural design. Verify spans, screw model, length and spacing against the ground-screw supplier's load tables and your actual ground conditions before building.